

SPI Serial Peripheral Interface

MOSI - Master Output Slave Input
MISO - Master Input Slave Output
CPOL - Clock Polarity
CPHA - Clock Phase

- ▷ Developed by **Motorola**, mid **1980s**
- ▷ Appeared in their **6855 custom 8 bit microcontrollers**

Synchronous :- It used **dedicated clock line**.

Simple : It requires **no complex addressing protocols** or hardware overhead

Fast : Operated in **full duplex**

- ▷ Serial Synchronous communication used to **connect a controller** to nearby **peripherals**.
- ▷ **Baseline 4 wire SPI**, one controller generates a **serial clock** and selects one peripheral using a **chip-select signal**.
- ▷ **Full duplex** : one bit is shifted from **controller to peripheral** and one bit is shifted from **peripheral to controller** on **each serial clock cycle**.

Terminology and Roles :-

Term	Meaning
Controller, host, main, master	Device that normally initiates the transfer, drives SCLK, and controls CS/SS.
Peripheral, target, subnode, slave	Device selected by the controller. It responds while its chip select is active.
SCLK, SCK, CLK	Serial clock generated by the controller in the baseline bus.
MOSI, COPI, SDI	Controller output, peripheral input.
MISO, CIPO, SDO	Controller input, peripheral output.
CS, SS, nCS, nSS	Chip select or slave select. Often active low.
Word	A group of serial bits shifted as one programmed unit, commonly 8 bits but not required to be 8.
Transfer	One contiguous set of serial clock edges, usually while CS is asserted.
Transaction	A higher-level operation such as command plus address plus dummy cycles plus data.

Signal overview :-

Baseline 4-wire SPI uses these signals:

Signal	Driven By	Typical Direction	Purpose
SCLK	Controller	Controller to all peripherals	Serial timing reference.
MOSI	Controller	Controller to selected peripheral	Command, address, write data, or dummy bits from controller.
MISO	Selected peripheral	Peripheral to controller	Read data, status, or dummy bits from peripheral.
CS/SS	Controller	One line per peripheral in common topology	Selects the active peripheral. Usually active low.

- ▷ When **no peripheral is selected**, **inactive peripherals should not drive the shared MISO line**.

- ▷ SPI is usually used for **short-distance communication**.

Applications:-

- ▷ **Serial NOR flash and EPROM**
- ▷ **ADC and DAC devices**
- ▷ **Sensors**
- ▷ **Display controllers**
- ▷ **Audio codecs**
- ▷ **Shift registers**

SPI Architecture :-

▷ SPI does not use address field on bus to choose a device.

▷ Device selection is normally done outside the data stream using a dedicated chip-select signal.

• SPI system has three parts:-

▷ SPI can have one main and can have one or multiple subnodes

Part	Role
Controller	Starts the transfer, drives SCLK, drives MOSI, and controls CS.
Peripheral	Responds only when selected, samples MOSI, and drives MISO.
Bus signals	Carry clock, data, and selection information between controller and peripheral.

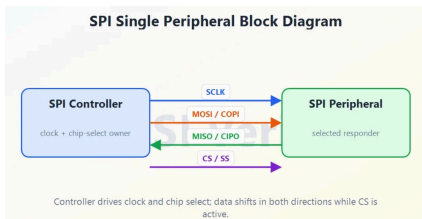
▷ Controller decides :-

- (1) decides when the transaction starts
- (2) how fast the clock runs
- (3) which clock edge is used for sampling
- (4) how many bits are transferred
- (5) when the selected peripheral is released.

▷ Peripheral :-

- (1) waits for CS to become active
- (2) follows the incoming SCLK
- (3) samples incoming bits on MOSI
- (4) Returns outgoing bits on MISO if transaction requires read data.

Single Peripheral :-



▷ Point-to-Point peripheral links

▷ SCLK tells both sides when each bit position occurs

▷ MOSI carries controller-to-peripheral information such as commands or write data

▷ MISO carries peripheral-to-controller information such as read data or status

▷ CS defines the active transaction window.

▷ When CS is inactive, the peripheral should ignore the SPI clock and data lines

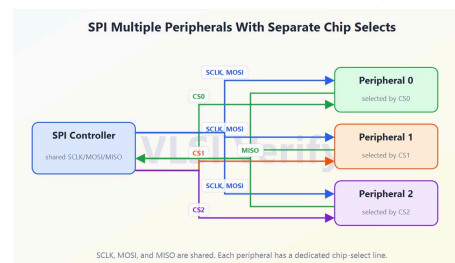
Multiple Peripherals :-

▷ SCLK, MOSI, MISO are shared.

▷ Each peripheral gets a separate chip-select line.

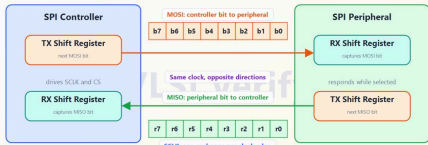
▷ Only one peripheral is normally selected at a time.

▷ Only the selected peripheral should drive MISO.



Full-duplex :-

SPI Full-Duplex Shift Register Idea



▷ SPI sends and receives at the same time

▷ Even when the controller only wants to read, it sends dummy bits

Transaction Vocabulary :-

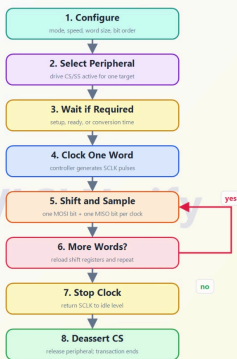
Term	Simple Meaning
Bit	One value shifted on MOSI or MISO during a clock cycle
Word	A group of bits shifted as a unit, commonly 8 bits but not always
Transfer	One continuous clocked movement of one or more words
Transaction	The complete selected operation framed by CS/SS
Dummy bits	Bits sent mainly to create clock cycles for receiving data
Phase	A logical part of a transaction, such as command, address, dummy, or data

Step-by-Step Flow :-

Step	What Happens	Why It Matters
1. Configure	Controller sets mode, clock speed, word size, bit order, and target peripheral	The controller and peripheral must agree on timing before bits are exchanged
2. Select peripheral	Controller drives CS/SS active for one peripheral	Only the selected peripheral should respond and drive MISO
3. Wait if required	Controller waits before clocking if the device needs setup, ready, or conversion time	Some peripherals need time after selection before valid clocking begins
4. Clock one word	Controller generates the required SCLK pulses for one word	No clock means no SPI data movement
5. Shift and sample	One bit moves on MOSI and one bit moves on MISO during each clock cycle	This is the full-duplex heart of SPI
6. More words?	Controller reloads/continues shifting if the transaction has more command, address, dummy, or data words	Multi-byte transactions remain inside the same selected window when required
7. Stop clock	Controller stops toggling SCLK and returns it to the idle level	The clocked portion of the exchange is complete
8. Deassert CS	Controller releases the peripheral	The transaction ends; many peripherals latch, commit, or reset parser state here

Transaction Flow Diagram :-

SPI Transaction Flow

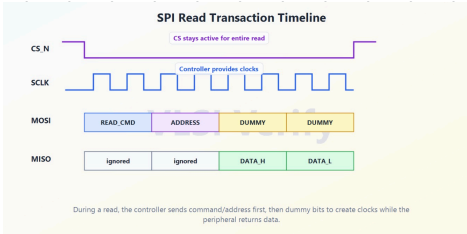


A transaction is a CS-framed exchange: select one peripheral, clock one or more words, then stop the clock and release CS.

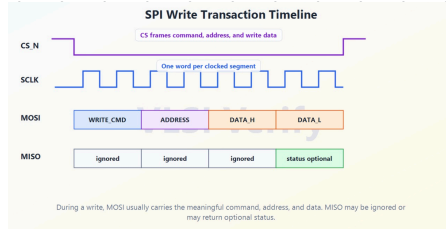
▷ During a clocked word,

- (1) The controller places or shifts the next output bit onto MOSI.
- (2) The peripheral places or shifts the next output bit onto MISO.
- (3) On the configured sampling edge, both sides capture the incoming bit.
- (4) The bit counter advances
- (5) The process repeats until the word is complete.

Read Transaction :-



Write Transaction :-



SPI Modes : define the clock idle level and the clock edge used for sampling data.

Table 1. SPI Modes with CPOL and CPHA

Parameter	Meaning
CPOL	Clock idle level
CPHA	Which clock edge samples data

Table 1. SPI Modes with CPOL and CPHA

SPI Mode	CPOL	CPHA	Clock Polarity in Idle State	Clock Phase Used to Sample and/or Shift the Data
0	0	0	Logic low	Data sampled on rising edge and shifted out on the falling edge
1	0	1	Logic low	Data sampled on the falling edge and shifted out on the rising edge
2	1	0	Logic high	Data sampled on the falling edge and shifted out on the rising edge
3	1	1	Logic high	Data sampled on the rising edge and shifted out on the falling edge

Examples :-

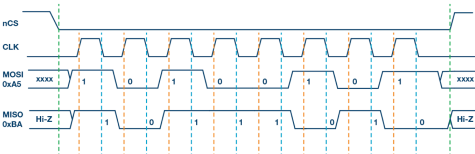
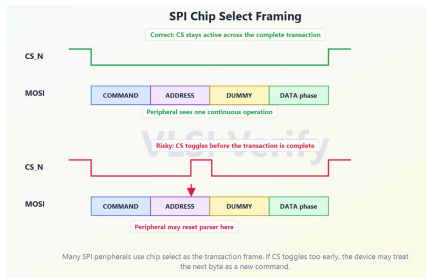
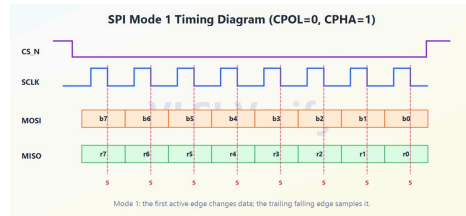
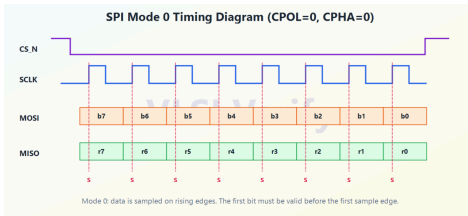


Figure 2. SPI Mode 0, CPOL = 0, CPHA = 0: CLK idle state = low, data sampled on rising edge and shifted on falling edge.

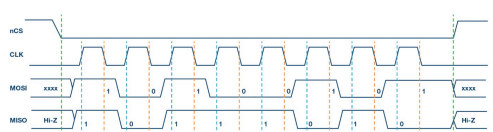


Figure 3. SPI Mode 1, CPOL = 0, CPHA = 1: CLK idle state = low, data sampled on the falling edge and shifted on the rising edge.

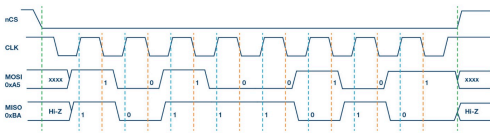


Figure 4. SPI Mode 2, CPOL = 1, CPHA = 0: CLK idle state = high, data sampled on the falling edge and shifted on the rising edge.

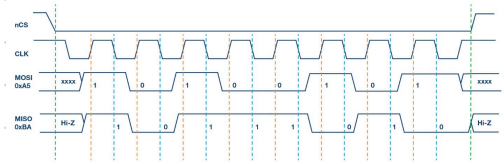


Figure 5. SPI Mode 3, CPOL = 1, CPHA = 1: CLK idle state = high, data sampled on the rising edge and shifted on the falling edge.

✓ Daisy-Chain Method:

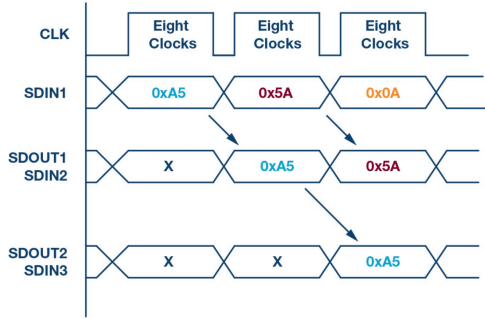
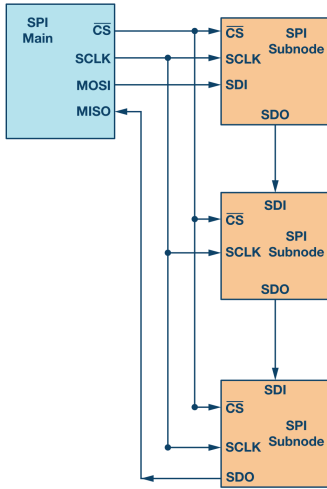


Figure 8. Daisy-chain configuration: data propagation.

References :

- 1) VLSI Verify
- 2) Analog Devices